MIL-C-11015/18D
28 September 1970
SUPERSEDING
MIL-C-11015/18C
24 May 1968

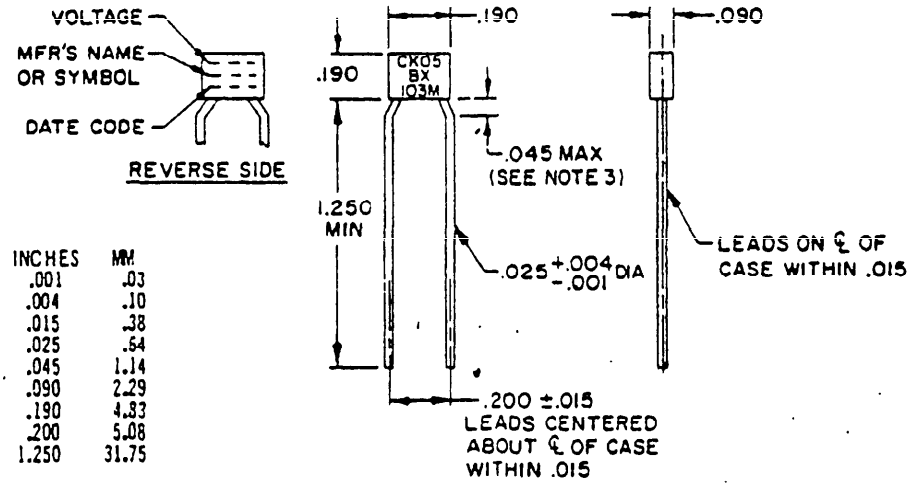
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), STYLE CK05

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-11015.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR NEW DESIGN
after 28 September 1970. Use MIL-C-39014/1



INCHES	MM
.001	.03
.004	.10
.015	.38
.025	.64
.045	1.14
.090	2.29
.190	4.83
.200	5.08
1.250	31.75

NOTES:
1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. For flush mounting, a .078 (1.98 mm) printed-circuit hole is required to clear shoulder.
4. Unless otherwise specified, tolerance is ±.010 (.25 mm).

FIGURE 1. Style CK05 capacitors.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1.
Case type - Flat plate, molded or pre-formed.
Capacitance value - See table I.
Capacitance tolerance - ±10 percent (K) or ±20 percent (M) as shown in table I.
Rated temperature - -55° to +125° C (although 10 pF through 1,000 pF capacitors are identified as B units, they shall have the capability of -55° to +150° C operation, see figure 2.)

Dielectric withstanding voltage:
Dielectric (DWV): Method 301 of MIL-STD-202.
Test voltage - 250 percent of rated voltage.
Body insulation: In accordance with MIL-C-11015.
Test potential - 1,300 volts dc.

Barometric pressure (reduced): Method 105 of MIL-STD-202. 0.82 inch of mercury (80,000 ft).
Test potential - 150 percent of rated voltage.

FSC 5910

Source: https://assist.dla.mil -- Downloaded: 20260-17T14:56Z
Check the source to verify that this is the current version before use.

TABLE I. Style CK05 characteristics.

Type designation 1/	Rated voltage	Rated temperature and voltage-temperature limits	Capacitance	Capacitance tolerance
	volts, dc		pF	
CK05BX100-	200	BX	10	K, M
CK05BX120K	200	BX	12	K
CK05BX150-	200	BX	15	K, M
CK05BX180K	200	BX	18	K
CK05BX220-	200	BX	22	K, M
CK05BX270K	200	BX	27	K
CK05BX330-	200	BX	33	K, M
CK05BX390K	200	BX	39	K
CK05BX470-	200	BX	47	K, M
CK05BX560K	200	BX	56	K
CK05BX680-	200	BX	68	K, M
CK05BX820K	200	BX	82	K
CK05BX101-	200	BX	100	K, M
CK05BX121K	200	BX	120	K
CK05BX151-	200	BX	150	K, M
CK05BX181K	200	BX	180	K
CK05BX221-	200	BX	220	K, M
CK05BX271K	200	BX	270	K
CK05BX331-	200	BX	330	K, M
CK05BX391K	200	BX	390	K
CK05BX471-	200	BX	470	K, M
CK05BX561K	200	BX	560	K
CK05BX681-	200	BX	680	K, M
CK05BX821K	200	BX	820	K
CK05BX102-	200	BX	1,000	K, M
CK05BX122K	100	BX	1,200	K
CK05BX152-	100	BX	1,500	K, M
CK05BX182K	100	BX	1,800	K
CK05BX222-	100	BX	2,200	K, M
CK05BX272K	100	BX	2,700	K
CK05BX332-	100	BX	3,300	K, M
CK05BX392K	100	BX	3,900	K
CK05BX472-	100	BX	4,700	K, M
CK05BX562K	100	BX	5,600	K
CK05BX682-	100	BX	6,800	K, M
CK05BX822K	100	BX	8,200	K
CK05BX103-	100	BX	10,000	K, M
CK05BX123K	50	BX	12,000	K
CK05BX153-	50	BX	15,000	K, M
CK05BX183K	50	BX	18,000	K
CK05BX223-	50	BX	22,000	K, M
CK05BX273K	50	BX	27,000	K
CK05BX333-	50	BX	33,000	K, M
CK05BX393K	50	BX	39,000	K
CK05BX473-	50	BX	47,000	K, M
CK05BX563K	50	BX	56,000	K
CK05BX683-	50	BX	68,000	K, M
CK05BX823K	50	BX	82,000	K
CK05BX104-	50	BX	100,000	K, M

1/ Where applicable, the complete type designation will include an additional symbol to indicate capacitance tolerance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:56Z
Check the source to verify that this is the current version before use.

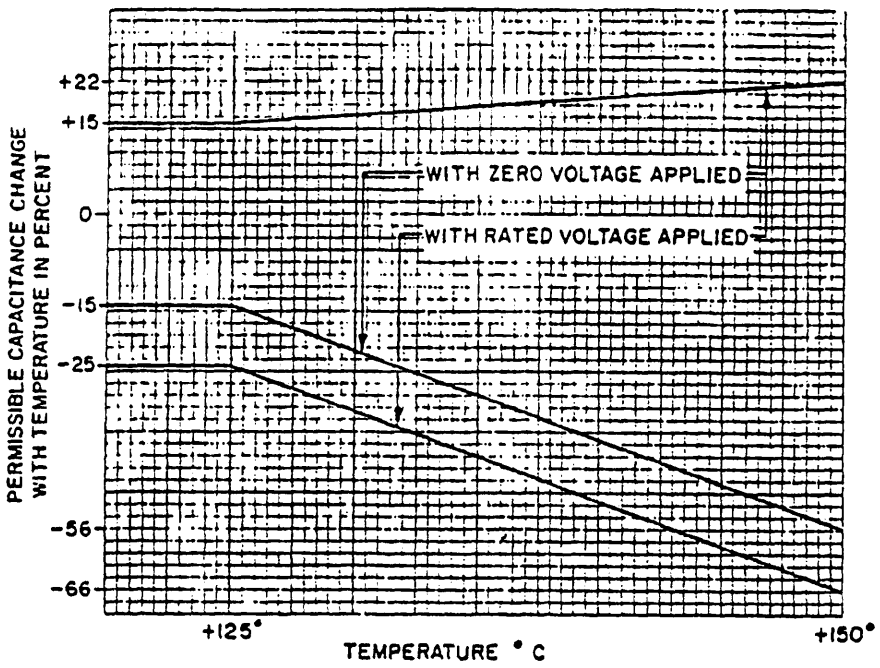


FIGURE 2. Extended voltage-temperature limits (10 pF through 1,000 pF capacitors).

Insulation resistance (IR): Method 302 of MIL-STD-202, rated voltage applied. 100,000 megohms, min or 1,000 megohm-microfarads, min, whichever is less.

Capacitance (Cap.): Method 305 of MIL-STD-202.
Test frequency:
100 pF or less - 1 MHz ± 1 kHz.
Over 100 pF - 1 kHz ± 100 Hz.
Test voltage - 1.0 ± 0.2 volt rms.

Dissipation factor (DF): 2.5 percent, max.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 G's).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G's).

Temperature cycling and immersion:
Temperature cycling: Method 102 of MIL-STD-202, condition D, except step 3 shall be at high test temperature.
Immersion: Method 104 of MIL-STD-202, condition . .
IR - Not less than 10 percent of initial requirement.

Salt spray (corrosion): Not applicable.

Terminal strength: In accordance with MIL-C-11015, 5 pounds applied.

Moisture resistance: Method 106 of MIL-STD-202.
DWV - 250 percent of rated voltage.
IR - Not less than 10 percent of initial requirement.
Cap. - Within tolerance of table I value.

Solderability: Method 208 of MIL-STD-202, 2 terminals.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.
IR - 100,000 megohms, min or 1,000 megohm-microfarads, min, whichever is less.
ΔCap. - ±5 percent of initial measurement.
ΔDF - 0.5 percent, max.

Voltage-temperature limits: In accordance with MIL-C-11015 and figure 2 (for 10 pF through 1,000 pF capacitors).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:56Z
Check the source to verify that this is the current version before use.

Life (at elevated ambient temperature):
Method 108 of MIL-STD-202. 2,000 hours (for qualification); 250 hours (performance check); 1,750 hours (continuation test).
Test potential - 200 percent of rated voltage.
IR - 1,000 megohms, min or 10 megohm-microfarads, min, whichever is less (at 125°C), and not less than 10 percent of initial requirement (at 25°C).
DF - 2.5 percent, max (at 25°C).

Marking: In accordance with MIL-C-11015, and as shown in figure 1.

Supersession data: Items having a "CW" voltage-temperature limit formerly covered by Rev. J and earlier issues, of this specification sheet are superseded as indicated in table II.

TABLE II. Cross-reference of substitute items.

Superseded item 1/	Recommended substitute item	Superseded item 1/	Recommended substitute item
CK05CW100-	CK05BX100-	CK05CW121K	CK05BX121K
CK05CW120K	CK05BX120K	CK05CW151-	CK05BX151-
CK05CW150-	CK05BX150-	CK05CW181K	CK05BX181K
CK05CW180K	CK05BX180K	CK05CW221-	CK05BX221-
CK05CW220-	CK05BX220-	CK05CW271K	CK05BX271K
CK05CW270K	CK05BX270K	CK05CW331-	CK05BX331-
CK05CW330-	CK05BX330-	CK05CW391K	CK05BX391K
CK05CW390K	CK05BX390K	CK05CW471-	CK05BX471-
CK05CW470-	CK05BX470-	CK05CW561K	CK05BX561K
CK05CW560K	CK05BX560K	CK05CW681-	CK05BX681-
CK05CW680-	CK05BX680-	CK05CW821K	CK05BX821K
CK05CW820K	CK05BX820K	CK05AW102-	CK05BX102-
CK05CW101-	CK05BX101-		

1/ Where required, the type designation will include comparable capacitance tolerance symbol.

Inactive styles: For new design it is recommended that units of MIL-C-39014/1 be considered as listed in table III. Although table III lists only the highest failure rate level, lower failure rate levels may be considered. For substitutability of the inactivated styles, capacitors in MIL-C-39014/1 of the same capacitance value, tolerance, rated voltage and temperature coefficient can be used regardless of the failure rate designation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:56Z
Check the source to verify that this is the current version before use.

TABLE III. Cross reference of inactivated styles.

From inactivated MIL-C-11015	To MIL-C-39014/1 L (2.0) M39014/01-	From inactivated MIL-C-11015	To MIL-C-39014/1 L (2.0) M39014/01-
CK05BX100K	0361	CK05BX102M	0398
CK05BX100M	0362	CK05BX122K	0399
CK05BX120K	0363	CK05BX152K	0400
CK05BX150K	0364	CK05BX152M	0401
CK05BX150M	0365	CK05BX182K	0402
CK05BX180K	0366	CK05BX222K	0403
CK05BX220K	0367	CK05BX222M	0404
CK05BX220M	0368	CK05BX272K	0405
CK05BX270K	0369	CK05BX332K	0406
CK05BX330K	0370	CK05BX332M	0407
CK05BX330M	0371	CK05BX392K	0408
CK05BX390K	0372	CK05BX472K	0409
CK05BX470K	0373	CK05BX472M	0410
CK05BX470M	0374	CK05BX562K	0411
CK05BX560K	0375	CK05BX682K	0412
CK05BX680K	0376	CK05BX682M	0413
CK05BX680M	0377	CK05BX822K	0414
CK05BX820K	0378	CK05BX103K	0415
CK05BX101K	0379	CK05BX103M	0416
CK05BX101M	0380	CK05BX123K	0417
CK05BX121K	0381	CK05BX153K	0418
CK05BX151K	0382	CK05BX153M	0419
CK05BX151M	0383	CK05BX183K	0420
CK05BX181K	0384	CK05BX223K	0421
CK05BX221K	0385	CK05BX223M	0422
CK05BX221M	0386	CK05BX273K	0423
CK05BX271K	0387	CK05BX333K	0424
CK05BX331K	0388	CL05BX333M	0425
CK05BX331M	0389	CK05BX393K	0426
CK05BX391K	0390	CK05BX473K	0427
CK05BX471K	0391	CK05BX473M	0428
CK05BX471M	0392	CK05BX563K	0429
CK05BX561K	0393	CK05BX683K	0430
CK05BX681K	0394	CK05BX683M	0431
CK05BX681M	0395	CK05BX823K	0432
CK05BX821K	0396	CK05BX104K	0433
CK05BX102K	0397	CK05BX104M	0434

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MI
Navy - AS, OS, SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5910-1003-1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:56Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND (ELEX 8111)
WASHINGTON, DC 20360

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:56Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION –	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER (Specify): _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYYYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:56Z
Check the source to verify that this is the current version before use.